

SPOKANE INTL, WA, USA

WMO: 727850

Lat: 47.622N

Lon: 117.528W

Elev: 2353

StdP: 13.49

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24157

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.1	11.8	-2.9	5.1	9.2	3.3	7.1	15.4	30.4	40.2	26.6	38.9	5.5	30	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	93.2	62.6	90.1	61.6	86.6	60.5	65.0	86.9	63.4	85.1	61.9	82.6	8.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.7	77.5	68.4	55.5	71.6	67.9	53.6	66.7	67.1	31.0	86.8	29.8	85.0	28.7	82.6	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	21.8	18.9	DB	-2.1	98.7	8.0	2.9	-7.8	100.7	-12.5	102.4	-17.0	104.0	-22.8	106.1	(4)
(5)			WB	-2.8	67.6	7.8	2.6	-8.4	69.5	-12.9	71.0	-17.3	72.5	-22.9	74.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.5	29.5	32.3	39.7	46.6	55.6	62.0	71.2	70.1	60.9	47.5	36.3	28.7		
	DBStd	16.48	8.39	7.91	6.53	6.51	7.58	7.13	6.59	6.43	7.40	7.01	7.77	7.97		
	HDD50	2837	636	496	321	144	30	3	0	0	7	128	412	660		
	HDD65	6539	1101	915	783	554	306	140	22	26	165	542	860	1125		
	CDD50	2276	0	0	3	41	203	361	659	623	332	52	2	0		
	CDD65	505	0	0	0	0	14	49	216	184	41	1	0	0		
	CDH74	6526	0	0	0	12	215	671	2781	2278	558	11	0	0		
	CDH80	2719	0	0	0	1	46	223	1278	991	179	1	0	0		
	WSAvg	8.4	7.8	8.3	9.6	9.6	8.9	9.0	8.2	7.8	7.6	8.0	8.2	7.8		
	PrecAvg	15.6	1.9	1.4	1.5	1.2	1.5	1.1	0.6	0.6	0.7	1.0	2.1	2.2		
(15) Precipitation	PrecMax	25.2	4.5	3.3	3.8	2.6	3.7	2.8	2.3	1.9	1.9	3.3	4.8	4.9		
	PrecMin	10.5	0.4	0.0	0.2	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.2	0.6		
	PrecStd	2.9	0.9	0.7	0.7	0.7	0.8	0.6	0.6	0.6	0.5	0.8	1.1	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	52.7	63.2	75.3	85.2	92.4	98.0	96.7	90.6	74.9	58.2	50.6		
		MCWB	43.5	45.5	48.6	54.8	58.9	62.8	64.7	63.3	60.8	56.0	52.1	46.2		
	2%	DB	44.2	48.5	58.0	69.0	80.0	85.9	94.0	92.9	85.2	67.9	53.2	44.9		
		MCWB	40.8	43.0	46.6	52.5	57.8	60.7	63.0	62.4	59.0	52.8	48.1	41.8		
	5%	DB	41.4	45.2	54.1	63.6	75.2	81.3	90.8	89.6	80.6	63.2	49.9	41.2		
		MCWB	39.0	40.6	44.9	49.5	55.4	59.1	61.9	61.1	57.8	51.1	45.1	38.9		
	10%	DB	39.1	42.2	50.5	59.1	70.5	77.2	87.5	85.9	75.9	58.9	47.0	38.4		
		MCWB	37.1	38.4	42.8	47.0	53.7	57.3	61.0	60.3	56.3	48.6	43.1	36.6		
	0.4%	WB	45.0	47.1	51.2	57.3	61.4	65.0	68.0	66.3	63.3	58.1	53.1	46.3		
		MCDB	46.9	50.8	58.8	71.4	78.5	87.0	89.3	89.6	82.3	69.8	56.2	49.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.4	43.9	48.1	53.3	59.0	62.5	65.6	64.3	61.0	54.7	48.7	42.0		
		MCDB	43.5	47.5	55.5	65.5	75.2	81.5	87.2	87.0	80.6	64.7	52.5	44.6		
	5%	WB	39.4	41.3	45.9	50.4	57.2	60.5	64.0	62.7	59.1	52.3	46.1	39.4		
		MCDB	41.0	44.5	52.6	62.0	72.0	77.8	85.9	84.9	77.3	61.2	49.0	40.8		
	10%	WB	37.3	38.9	43.9	48.1	55.1	58.8	62.3	61.3	57.2	50.0	43.8	36.7		
		MCDB	38.7	41.7	50.3	57.5	67.9	74.8	83.5	82.7	73.5	57.3	46.8	38.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	12.7	16.3	19.1	21.2	22.0	26.2	26.0	23.8	18.6	12.0	9.1		
		MCDBR	10.9	15.5	21.8	25.6	27.5	27.8	30.1	30.3	30.2	24.3	14.6	10.6		
		MCWBR	8.3	10.4	12.9	12.8	11.1	10.1	10.3	11.1	12.1	12.5	10.0	8.8		
		MCDBR	10.3	13.6	19.1	23.5	24.3	25.5	27.7	27.7	27.2	21.4	12.8	10.0		
	5% WB	MCWBR	8.3	9.8	12.3	12.5	10.5	9.7	10.2	10.5	11.5	11.6	9.5	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.280	0.294	0.312	0.325	0.327	0.327	0.337	0.314	0.300	0.280	0.268		
	taud		2.499	2.539	2.555	2.490	2.472	2.496	2.496	2.452	2.507	2.544	2.556	2.501		
	Ebn at Noon		264	286	298	300	298	296	295	287	286	274	259	251		
	Edh at Noon		20	23	27	31	33	33	32	32	28	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		331	626	960	1438	1848	1990	2340	1943	1384	821	390	260		
	RadStd		42	85	113	138	148	174	128	128	131	92	43	35		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+92
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	-0.72	N/A	N/A	N/A	+171	+79

Nomenclature: See separate page